

☆ Information Theory

* Information:

⇒ Information associated with any event is inversely proportional to probability of occurrence.

$$\Rightarrow I \propto \frac{1}{p}$$

$$\Rightarrow \boxed{I = \log_2 \frac{1}{p} \text{ bits}} \Leftarrow \text{H.B.}$$

* Entropy:

⇒ The average amount of information is called Entropy.

$$\boxed{H = \sum_i p_i \log_2 \frac{1}{p_i} \text{ bits/symbol}} \Leftarrow \text{H.B.}$$

⇒ Information and Entropy can be understood by following example.

Q A discrete source generates 4 symbols x_1, x_2, x_3, x_4 with probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}$ respectively. Calculate the entropy. Calculate the entropy when the symbols occur with equal probability.

$$\Rightarrow H = \sum_i P_i \log_2 \frac{1}{P_i}$$

$$\therefore H = P_1 \log_2 \frac{1}{P_1} + P_2 \log_2 \frac{1}{P_2} + P_3 \log_2 \frac{1}{P_3} + P_4 \log_2 \frac{1}{P_4}$$

$$\therefore H = P_4 \log_2 \frac{1}{P_4}$$

$$\Rightarrow H = \frac{1}{2} \log_2 2 + \frac{1}{4} \log_2 4 + \frac{1}{8} \log_2 8 + \frac{1}{8} \log_2 8$$

$$H = \frac{1}{2} (1) + \frac{2}{4} + \frac{3}{8} + \frac{3}{8}$$

$$\Rightarrow \boxed{H = 1.75 \text{ bits/symbol}}$$

$\Rightarrow$ If x_1, x_2, x_3, x_4 have equal probability,

then $H = 4 \left(\frac{1}{4} \log_2 4 \right)$

$$\therefore \boxed{H = 2 \text{ bits/symbol}}$$

$\Rightarrow$ Information Theory Concept is used to transmit non-electric discrete signals through a digital communication.

$\Rightarrow$ There are two types of coding are used. It is called Source Coding.

$\rightarrow$ Fixed length coding.

⇒ In Early days morse code is used to transmit data. i.e. $\cdot$ & $-$ are transmitted as a data.

① Fixed length Coding:

⇒ $\left\{ \begin{array}{l} A \quad 00000 \\ B \quad 00001 \\ C \quad \vdots \\ \vdots \quad \vdots \\ X \quad \vdots \\ Z \quad \vdots \end{array} \right.$ 26 alphabets
So, $2^n = 26$
⇒ $n = 5$ bits required for each symbol.

⇒ In fixed length Coding, Code length is fixed for every Symbol. i.e. code is independent of the probability of occurrence of that symbol.

② Variable Length Coding:

⇒ In Variable Length Coding, Length of code is not fixed. It is depend on the probability of occurrence of that code.

Let, 4 symbols x_1, x_2, x_3, x_4 having prob. $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}$ respectively.

x_1	$\frac{1}{2}$	0
x_2	$\frac{1}{4}$	10

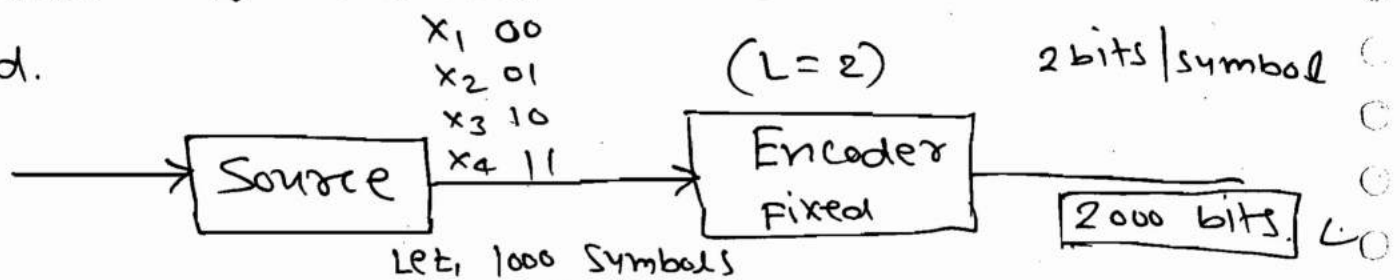
⇒ First arrange the prob. in ascending order.

⇒ Now, draw a line in a such a way that the sum of prob. of symbols upper to the line is exactly equal to the sum of prob. of symbols below the line.

⇒ Now, give code '0' to the upper symbols and '1' to the lower symbols.

⇒ Now, ~~take~~ the symbol which is have unique symbol leave it and Repeat process for the remaining symbol until unless we get the unique code for each and every symbol.

⇒ Entropy indicates the average no. of bits required to encode the each symbol if Variable Length Coding is used.



x_1	0	500	$\Rightarrow 500 \times 1 = 500$ bits
x_2	10	250	$\Rightarrow 250 \times 2 = 500$ bits
x_3	110	125	$\Rightarrow 125 \times 3 = 375$ bits
x_4	111	125	$\Rightarrow 125 \times 3 = 375$ bits

* Information rate & Bit rate.

$$\Rightarrow \text{Bit rate} = \frac{\text{bits}}{\text{sec}} = \frac{\text{bits}}{\text{symbol}} \times \frac{\text{symbol}}{\text{sec}}$$

$\downarrow$ Entropy $\downarrow$ Symbol rate.

$$\Rightarrow R_b = [E \times \text{Symbol rate}] \text{ bits/sec.}$$

$$\therefore R_b = \text{Entropy} \times \text{Symbol rate. bits/sec.}$$

* Efficiency of Code:

$$\eta = \frac{H}{\bar{L}}, \quad \bar{L} = \sum P_i L_i$$

$\Rightarrow$ for previous example,

$$H = 1.75 \text{ bits/symbol}$$

$$\bar{L} = \sum_i P_i L_i = \left(\frac{1}{2} \times 1\right) + \left(\frac{1}{4} \times 2\right) + \left(\frac{1}{8} \times 3\right) + \left(\frac{1}{8} \times 3\right)$$

$$\bar{L} = 1.75 \text{ bits/symbol.}$$

$$\therefore \eta = \frac{1.75}{1.75}$$

$$\Rightarrow \boxed{\eta = 100\%}$$

$\Rightarrow$ But in all case, it is not possible that $\eta = 100\%$ as it is not possible to divide the symbol above and below line with equal probability.

Q
GATE - 2013

Consider two discrete source x & y each generating the symbol 1 & -1 with equal probability.

Let, $z = x + y$. Determine the entropy of z .

Solⁿ:

x	y	$z = x + y$	P
1	1	$\underline{2}$	$\underline{=}$
-1	-1	(1+1) 2	$\frac{1}{4}$
		(1-1, -1+1) 0	$\frac{1}{2}$
		(-1, -1) -2	$\frac{1}{4}$

$$\therefore \text{Sol } H = P \sum_i P_i \log_2 \frac{1}{P_i}$$

$$\therefore H = \frac{1}{4} \log_2 4 + \frac{1}{2} \log_2 2 + \frac{1}{4} \log_2 4.$$

$$\therefore H = \frac{2}{4} + \frac{1}{2} + \frac{2}{4}.$$

$$\therefore H = 3/2$$

$$\therefore H = 1.5 \text{ bits/symbols.}$$

Q A discrete source generate 3 symbols x_1, x_2, x_3 with Prob. $\frac{1}{2}, \frac{1}{4}, \frac{1}{4}$. The symbol rate is 3000 symbols/sec. Determine the information rate.

Soln:

$$H = \frac{1}{2} \log_2 2 + \frac{1}{4} \log_2 4 + \frac{1}{4} \log_2 4$$

$$H = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = 1.5 \text{ bits/symbols.}$$

$$\therefore R_b = [\text{Entropy} \times \text{symbol rate}] \text{ bits/sec.}$$

$$\therefore R_b = 1.5 \times 3000$$

$$= 4500 \text{ bits/sec}$$

$$\therefore \boxed{R_b = 4.5 \text{ Kbps}}$$

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$\Rightarrow$ Consider a discrete source generating a two symbol x_1 & x_2 .

$$x_1 \rightarrow P \quad \frac{1}{2}$$

$$x_2 \rightarrow (1-P) \quad \frac{1}{2}$$

$$\therefore H = \frac{1}{P} \log_2 P + \frac{1}{1-P} \log_2 (1-P).$$

for max H,

$$\frac{dH}{dP} = 0 \Rightarrow \boxed{P = \frac{1}{2}}$$



$\Rightarrow$ The entropy will be max. only when the symbol occurs with equal probability.

$$\begin{matrix} \uparrow \\ H_B \\ = \end{matrix} \rightarrow \therefore \boxed{H_{\max} = \log_2 M.}$$

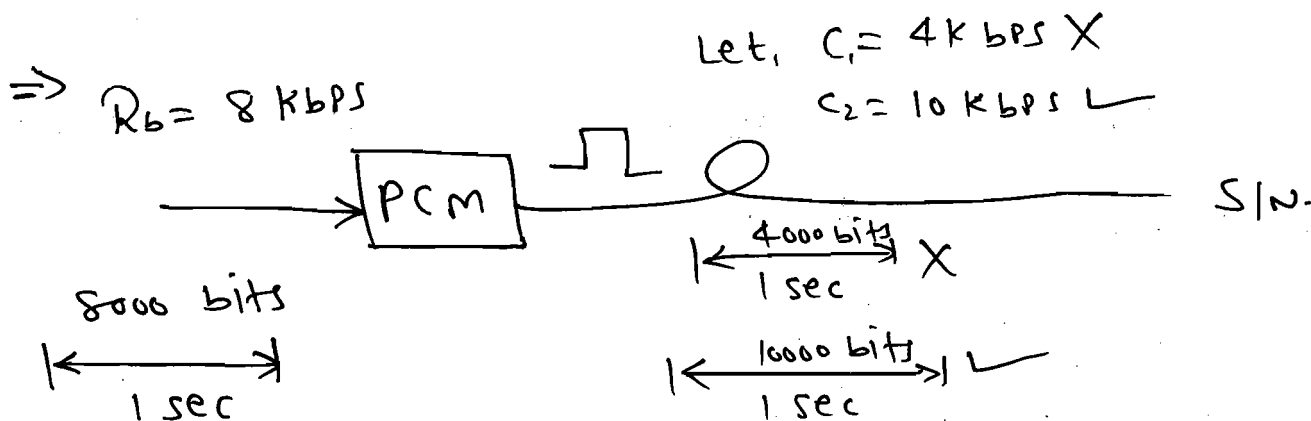
$$M=2 \Rightarrow H_{\max} = 1$$

$M = \text{no. of symbols.}$

$$M=4 \Rightarrow H_{\max} = \log_2 4 = 2.$$

☆ Channel Capacity:

Defⁿ: $\Rightarrow$ Channel Capacity is defined as the max. no of bits that the channel is capable of transmitting without any error.



$$\therefore C < R_b \text{ errors}$$

$$C \geq R_b \rightarrow \text{No errors.}$$

$\Rightarrow$ To determine the channel capacity

Shannon's Law is used,

i.e.

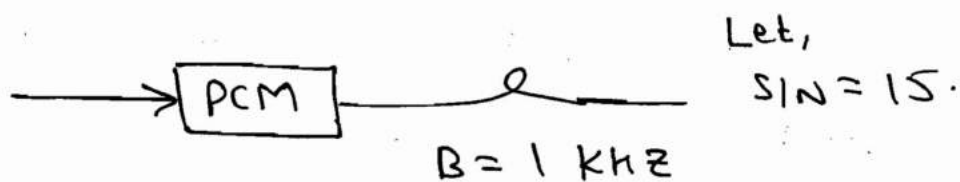
linear scale.

$C = B \log_2 [1 + S/N] \quad \text{bits/sec.}$

⇒ Channel Capacity is directly proportional to the channel BW.

⇒ Channel Capacity is also called data transfer rate.

e.g. :



① $B = 1 \text{ kHz}$

⇒ $C = B \log_2 (1 + S/N)$

$C = 10^3 \log_2 (1 + 15)$

∴ $C = 4 \text{ Kbps.}$

Twisted Pair

② $B = 1 \text{ MHz.}$

⇒ $C = 10^6 \log_2 (1 + 15)$

$C = 4 \text{ Mbps.}$

Coaxial Cable

③ $B = 1 \text{ GHz.}$

⇒ $C = 4 \text{ Gbps}$

Foc.

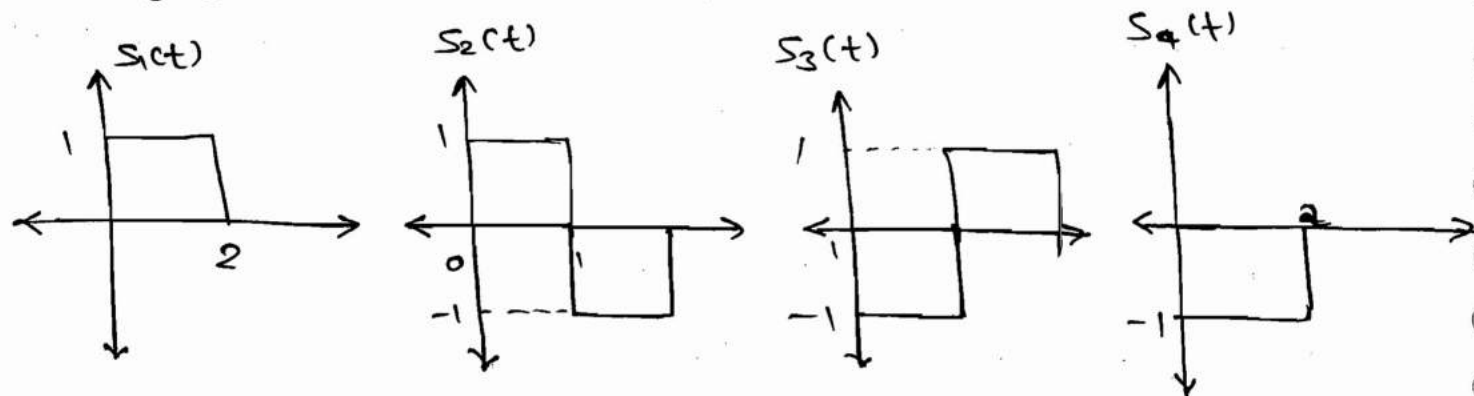
⇒ When $B \rightarrow \infty \Rightarrow$

$C_{\text{max}} = 1.44 \frac{S}{\text{No.}}$

signal power.

* Constellation diagram Concept:

⇒ Consider the four signals as shown in figure.



⇒ The above four signals can be represented as a linear combination of orthonormal basis ϕ^n .

⇒ The orthonormal basis function should satisfy the following two properties:

① The energy should be 1.

$$\int_{-\infty}^{\infty} \phi_1(t) dt = 1, \quad \int_{-\infty}^{\infty} \phi_2(t) dt = 1.$$

② They should be orthogonal to each other.

⇒ So, for this case $\phi_1(t)$ & $\phi_2(t)$ are as follow:



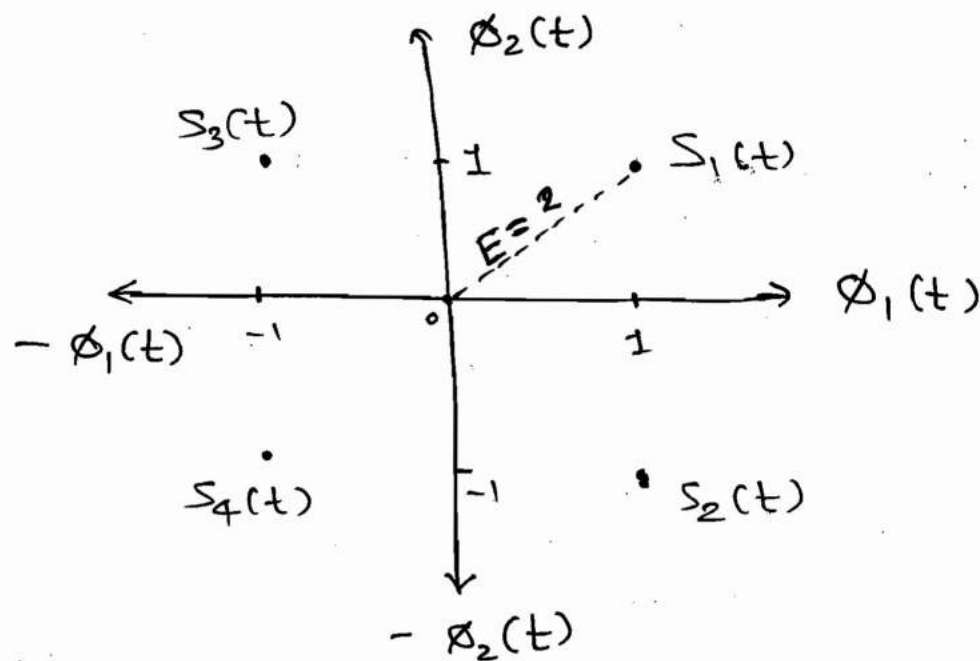
$$\Rightarrow \text{So, } S_1(t) = 1 \cdot \phi_1(t) + 1 \cdot \phi_2(t).$$

$$S_2(t) = 1 \cdot \phi_1(t) - 1 \cdot \phi_2(t).$$

$$S_3(t) = -1 \cdot \phi_1(t) + 1 \cdot \phi_2(t).$$

$$S_4(t) = -1 \cdot \phi_1(t) - 1 \cdot \phi_2(t).$$

$\Rightarrow$



(Signal space diagram (or)
Constellation diagram).

$\Rightarrow$ In, general $E=1$.

so, the signal $s(t)$ should be

$$\boxed{\sqrt{\frac{2}{N T_b}} \cdot \cos 2\pi f_c t \Rightarrow E=1} \Leftarrow \text{H.B.}$$

$$\text{as, } E = \frac{A^2 T_b}{2} = 1$$

$$A = \sqrt{\frac{2}{N T_b}} \Rightarrow A_c \cos 2\pi f_c t$$

$$= \sqrt{\frac{2}{N T_b}} \cdot \cos 2\pi f_c t$$

$$\left. \begin{aligned} s_{01}(t) &= 1 \cdot \sqrt{\frac{2}{T_b}} \cdot \cos 2\pi f_c t \Rightarrow E=1 \\ \phi_{21}(t) &= 1 \cdot \sqrt{\frac{2}{T_b}} \cdot \sin 2\pi f_c t \Rightarrow E=1 \end{aligned} \right\} \text{Standard form.}$$

$$\Rightarrow \underline{\text{ASK:}} \quad \begin{aligned} S_1(t) &= A_c \cos 2\pi f_c t & '1' \\ &= 0 & '0' \end{aligned}$$

$$\begin{aligned} E &= \int_0^{T_b} \frac{A_c^2}{|s_1(t)|^2} dt \Rightarrow E_b = P \cdot T_b \\ E_b &= \frac{A_c^2}{2} \cdot T_b \\ A_c &= \sqrt{\frac{2 E_b}{T_b}} \end{aligned}$$

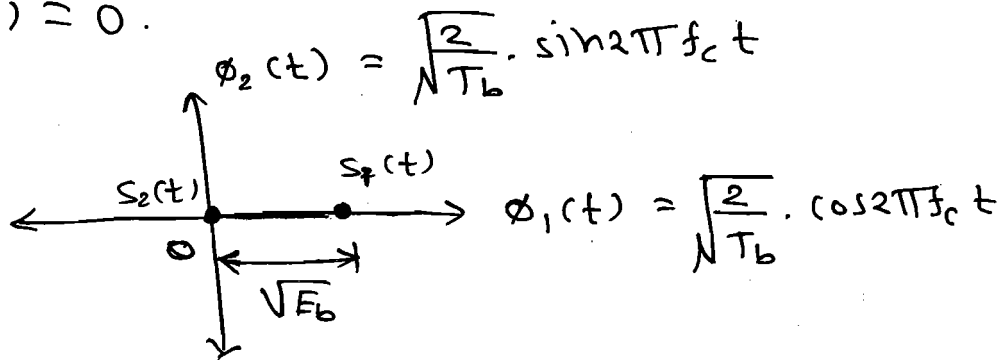
$$\therefore S_1(t) = \sqrt{\frac{2 E_b}{T_b}} \cdot \cos 2\pi f_c t \quad '1'$$

$$S_2(t) = 0 \quad '0'$$

$$\Rightarrow S_1(t) = \sqrt{E_b} \cdot \sqrt{\frac{2}{T_b}} \cdot \cos 2\pi f_c t$$

$$S_2(t) = 0$$

$\Rightarrow$



$\Rightarrow \phi_1(t)$ & $\phi_2(t)$ in a such way that

$$E=1. \quad \boxed{s_{01}(t) = \sqrt{\frac{2}{T_b}} \cdot \cos 2\pi f_c t}$$

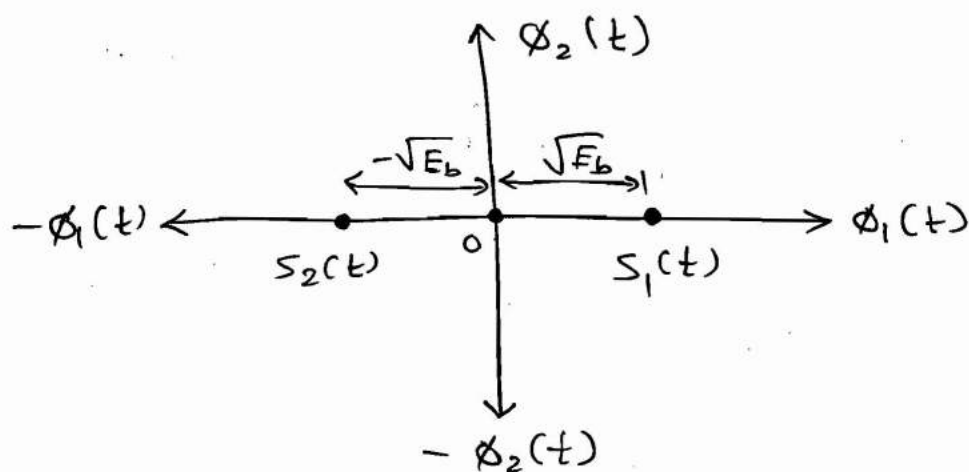
$\Rightarrow$ For BPSK:

$$S_1(t) = +\sqrt{E_b} \cdot \sqrt{\frac{2}{T_b}} \cdot \cos 2\pi f_c t \quad '1'$$

$$S_2(t) = -\sqrt{E_b} \cdot \sqrt{\frac{2}{T_b}} \cdot \cos 2\pi f_c t \quad '0'$$

$$\Rightarrow S_1(t) = +\sqrt{E_b} \cdot \phi_1(t) + 0 \cdot \phi_2(t)$$

$$S_2(t) = -\sqrt{E_b} \cdot \phi_1(t) + 0 \cdot \phi_2(t)$$



$\Rightarrow$ For QPSK:

$$S_1(t) = +\sqrt{E_b} \cdot \phi_1(t) + \sqrt{E_b} \phi_2(t) \Rightarrow 11$$

$$S_2(t) = +\sqrt{E_b} \cdot \phi_1(t) - \sqrt{E_b} \phi_2(t) \Rightarrow 10$$

$$S_3(t) = -\sqrt{E_b} \cdot \phi_1(t) + \sqrt{E_b} \phi_2(t) \Rightarrow 01$$

$$S_4(t) = -\sqrt{E_b} \phi_1(t) - \sqrt{E_b} \phi_2(t) \Rightarrow 00$$

